



Deep learning techniques for otologic imaging: a systematic review of segmentation methods

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Abstract

Developing segmentation models for otologic structures in medical images is fundamental for automated analysis and supporting clinical decision-making in ear-related patient care and surgical management. However, this task remains highly challenging because the ear contains small and morphologically complex substructures that often exhibit low visual contrast against surrounding tissues, making manual annotation labor-intensive and prone to inconsistency. Recent advances in deep learning (DL) have transformed medical image analysis across multiple clinical specialties, presenting new architectures to overcome these challenges. Beyond conventional convolutional neural networks (CNN), state-of-the-art approaches include vision transformers, diffusion models, and large-scale foundation models capable of capturing long-range dependencies, modelling sub-millimeter anatomy, and generalizing across variable imaging conditions. This systematic review provides the most comprehensive evaluation to date of how these DL models have been applied to otologic segmentation. Although DL has been widely reviewed in broader medical imaging, dedicated coverage of otologic applications is limited. Existing otologic reviews are typically narrow in scope, often confined to the middle ear, and overlook many clinically relevant anatomical domains. They also primarily emphasize CNN-based architectures and do not incorporate recent advances from 2023 to 2025, including transformer-based, diffusion-based, and hybrid architectures that integrate domain-prior strategies for enhanced contextual modelling. To address this gap, our review synthesizes research across all major otologic substructures and traces the evolution of DL architectures from CNN-based to hybrid and emerging models. We identify key performance trends and highlight persistent barriers to clinical translation, including severe class imbalance, limited multi-center datasets, variability in imaging, heterogeneous annotation standards, and high computational cost that restrict deployment in hospital settings. Based on these insights, we propose forward-looking research priorities to accelerate translation into clinical workflows. These include explainable and trustworthy DL models to support clinician confidence and regulatory approval, lightweight architectures suitable for diverse resource environments, domain adaptation across scanners and institutions, and scalable deployment frameworks compatible with diagnostic and surgical navigation systems. By integrating current advances, limitations, and future opportunities, this review presents a structured roadmap for

developing clinically deployable DL models for otologic segmentation and outlines how these technologies can enhance future diagnostic and surgical practice.

Keywords Otologic structure · Deep learning · Medical image segmentation · Convolutional neural network · Vision transformer · Diffusion

1 Introduction

The human ear is a vital sensory organ responsible for hearing and balance, playing a crucial role in environmental awareness and spatial orientation (Benson and Lane 2022; Lingam et al. 2019). It is anatomically categorized into three distinct regions: the outer ear, middle ear, and inner ear, each with unique structures and functional significance. As illustrated in Fig. 1, the outer ear consists of the auricle and the external auditory canal, which collect sound waves and direct them toward the tympanic membrane (eardrum). The middle ear contains three tiny ossicles: the malleus, incus, and stapes which amplify vibrations from the eardrum to the oval window of the cochlea. The inner ear includes the cochlea and vestibular system. While the cochlea transforms mechanical sound vibrations into neural signals for auditory perception, the vestibular system, composed of the semicircular canals, is essential for balance and spatial orientation.

Otologic diseases encompass a broad spectrum of conditions, including otitis media, cholesteatoma, otosclerosis, sensorineural hearing loss, vestibular disorders, facial nerve paralysis, and temporal bone fractures (Bassiouni et al. 2023). Accurate segmentation of otologic structures is essential for diagnosis, treatment planning, and surgical intervention. Procedures such as cochlear implantation, mastoidectomy, and tumor resection rely heavily on the detailed visualization of small and complex otologic substructures (Li et al. 2023a). High-resolution computed tomography (HRCT), typically acquired using multi-slice CT (MSCT) systems, is the imaging modality of choice for otologic evaluation due to its ability to provide high-contrast representation of intricate anatomical details (Pham et al. 2022; Natour et al. 2024) and to facilitate three-dimensional (3D) reconstruction of otologic anatomy. However, interpreting these high-resolution volumetric data is challenging and requires a high level of expertise due to fine-scale, spatially interconnected nature of otologic structures and their anatomical variability of otologic structures especially in pathological cases (Lee et al. 2024). These complexities underscore the urgent need for scalable and precise automated segmentation tools to support clinical workflows.

Deep learning (DL) has emerged as a powerful approach for otologic structure segmentation. DL-based methods offer consistent, objective, and automated analysis, thereby reducing human error and inter-observer variability (Powell et al. 2019; Nogovitsyn et al. 2019). Among these techniques, convolutional neural networks (CNNs) have demonstrated promising performance in otologic imaging tasks (Li et al. 2021; Li et al. 2023b). CNNs leverage hierarchical feature extraction, enabling the accurate identification of different anatomical structures at varying spatial scales (Hasan et al. 2023). Representative encoder-decoder models with skip connections effectively preserve local features and enhance structural delineation. When trained on large-scale annotated datasets, CNN-based models can achieve both reliable accuracy and robustness in clinical applications.

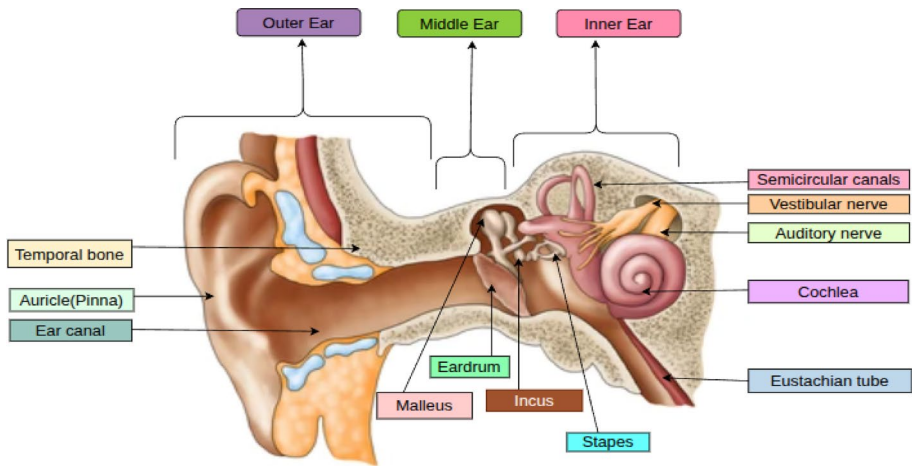


Fig. 1 An illustration of the anatomical structure of the human ear

Over time, various extensions of CNNs have been proposed to enhance feature representation and segmentation performance (Zhou et al. 2019; Niyas et al. 2022). Despite their advantages, CNNs often encounter challenges in handling boundary ambiguity and fine-grained detail recognition. These issues are primarily attributable to the progressive loss of spatial resolution introduced by successive down sampling operations during feature encoding, which diminishes the preservation of localized pixel-level features. This degradation reduces the network's capacity to capture sharp structural boundaries and fine-grained anatomical details, thereby exacerbating boundary ambiguity and compromising segmentation precision. As a result, CNNs may struggle to accurately delineate subtle boundaries between adjacent structures and capture small, intricate details. Furthermore, the inherently limited receptive fields of CNNs restrict their ability to learn long-range dependencies and global context within an image, which are essential for understanding spatial arrangements among distant anatomical components. Given that the ear comprises numerous small, spatially interconnected substructures, these limitations indicate that CNN-based models alone may be insufficient for optimal otologic structure segmentation.

There is an increasing demand for more advanced architectures for otologic structure segmentation. In response, transformer-based architecture has gained prominence as a powerful alternative (Strudel et al. 2021; Liu et al. 2021; Dosovitskiy et al. 2020). These models employ self-attention mechanisms, which allow each image patch to attend to all other elements in the input, thereby enabling the learning of global contextual relationships across the entire image. This effectively compensates for the limited local receptive fields of CNNs. Furthermore, hybrid approaches that integrate CNNs and transformers have been explored (Wang et al. 2021b; Iqbal et al. 2025; Tang et al. 2024; Lan et al. 2026; Wu et al. 2024; Guo et al. 2024). These models combine the strengths of both architectures by leveraging fine-grained local features and global contextual awareness. In addition, emerging architecture based on the diffusion process (Xie et al. 2025) and graph neural networks (Li et al. 2025b) have been investigated to further address the limitations of CNN-based approaches. Diffusion-based models mitigate the heavy reliance on large, annotated datasets by generating high-quality pseudo-labels from unlabeled data through iterative denoising, which are then

combined with real labels for joint supervised–unsupervised training. This enables effective learning of fine anatomical details even in data-scarce scenarios. Graph neural networks, on the other hand, explicitly encode spatial relationships between anatomical landmarks as graph priors, allowing efficient context propagation and reducing computational overhead compared to dense pixel-wise processing. These capabilities make both diffusion-based and graph-based approaches valuable complements to traditional CNNs, enhancing robustness, scalability, and clinical applicability in otologic segmentation.

Numerous surveys have been conducted on DL models for medical image analysis tasks (Ali et al. 2024; Minaee et al. 2021; Ali et al. 2025); however, comprehensive reviews focusing specifically on otologic segmentation remain scarce and are often limited in scope. Existing reviews tend to emphasize only a subset of otologic anatomy, particularly the middle ear (Ross et al. 2024), while neglecting the broader anatomical structures of the ear, despite their clinical significance. Furthermore, this survey primarily focuses on CNN-based approaches and includes a limited number of studies (only 10 from 2019 to 2023), thereby overlooking recent advancements in other emerging architectures and hybrid models that have shown considerable promise in medical image segmentation (Gao et al. 2025; Gao et al. 2021; Zhang et al. 2021; Manzari et al. 2024; Liu et al. 2025a; Wang et al. 2022c; Fu et al. 2024). In addition, discussions of experimental setups, covering target anatomical structures; clinical dataset attributes such as image modalities, image resolution, number of patients, and image dimensions; as well as evaluation metrics, backbone architecture, and key technical innovations, are generally superficial and fail to adequately reflect their critical role in assessing model performance and clinical applicability in otologic practice.

Motivated by the gaps, this review presents a systematic and comprehensive evaluation of DL-based segmentation models that encompass substructures across the entire ear anatomy. Most importantly, we review cutting-edge models developed between 2020 and 2025, with a focus on those integrating transformer architectures, diffusion mechanisms, and graph neural networks into hybrid frameworks capable of addressing key limitations of CNN-only approaches. By discussing their core technological innovations, current limitations, and clinical challenges in otologic applications, this review aims to guide future research directions. Ultimately, this review seeks to promote the development and clinical implementation of robust decision-making support systems that enhance diagnostic accuracy and facilitate surgical planning in otologic practice, thereby contributing to improved patient care and outcomes. The key contributions of this systematic review are summarized as follows:

- A total of 19 peer-reviewed studies published between 2020 and 2025 were systematically selected from an initial pool of 1,615 records in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines. This review presents a comprehensive evaluation of DL-based segmentation models targeting the entire ear anatomy; it includes more than ten critical substructures: the cochlea, sigmoid sinus, ossicular chain, oval window, round window, helicotrema, incus, stapes, malleus, facial nerve, semicircular canals, and bony labyrinth. Additionally, the selected study spans diverse imaging modalities, with dataset sizes reaching up to 325 patient cases.
- The evolution of otologic structure segmentation models is analyzed, from conventional CNNs to advanced hybrid approaches that incorporate transformer architectures, diffu-

sion mechanisms, and graph neural networks. Each methodological innovation is rigorously examined in terms of its ability to address both technical and clinical challenges in otologic structure segmentation.

- Promising future research directions are outlined to enhance the clinical viability of otologic structure segmentation models. Key enhancement areas include the development of lightweight, deployment-ready systems; the creation of large-scale, multi-institutional datasets; the integration of multi-modal imaging data; and the adoption of annotation-efficient training strategies. Special emphasis is placed on enabling efficient, precise training and improving model generalizability.

The remainder of this paper is organized as follows: Sect. 2 details the methodology used for this systematic review; Sect. 3 describes DL-based models for automated otologic structure segmentation in terms of architectures, key innovations, experiments, as well as associated challenges and limitations; Sect. 4 presents the discussion and key insights of this research; and Sect. 5 provides the overall conclusion of this review and outlines future research directions and opportunities for advancing otologic structure segmentation.

2 Methodology

The organization of this systematic review comprises the following components: (1) literature search strategy, (2) inclusion and exclusion criteria, and (3) literature search and study selection. The review was conducted by five investigators with varying levels of academic and research experience: two Ph.D. candidates, a recent Ph.D. graduate, and two associate professors. We followed the PRISMA 2020 recommendations appropriate for systematic reviews (Page et al. 2021), rather than the PRISMA-ScR extension for scoping reviews (Tricco et al. 2018). The PRISMA 2020 checklist and flow diagram were used to document literature identification, screening, eligibility, and inclusion processes in a standardized manner, consistent with other systematic reviews in medical imaging and artificial intelligence research (O'Shea et al. 2021; Abdullah 2025; AbuHaweeleh et al. 2025; Arar et al. 2025).

2.1 Literature search strategy

A comprehensive literature search was conducted across five major bibliographic databases, with Google Scholar used as a supplementary search source, to identify studies on otologic structure segmentation models. These databases broadly cover topics related to medical imaging technologies, artificial intelligence, and DL-based models. The search process began with Web of Science, a leading repository for DL and computer vision research, followed by PubMed, which emphasizes biomedical literature. To further broaden the scope of medical and life sciences literature, Embase was consulted. IEEE Xplore was also included due to its focus on medical imaging technologies. Next, Scopus, known for its multidisciplinary indexing, was explored. Google Scholar was used only as a supplementary search source to identify potentially relevant peer-reviewed journal articles that might not have been captured through the other bibliographic databases. Although Google Scholar may

index non-peer-reviewed records, all records retrieved from Google Scholar were screened according to the predefined eligibility criteria. Only peer-reviewed journal articles were retained for final inclusion. The initial search was limited to studies published between 11 February 2020 and 19 March 2025, and peer-review status and article type were verified during screening and eligibility assessment.

We selected two keyword groups, medical and technical terms, to ensure adequate domain coverage. The medical terms reflected the otologic and anatomical scope of the review, including “temporal bone”; “ear,” “inner ear,” “middle ear,” “ossicles,” “malleus,” “incus,” “stapes,” “cochlea,” and “facial nerve”. The technical terms covered core concepts in medical image segmentation and deep learning as well as factors relevant to small, complex otologic structures in CT or MRI, including medical image segmentation; CNNs; transformer networks; U-Net; attention mechanisms; dice similarity coefficient (DSC); multi-scale feature extraction; small object segmentation. Search strings were constructed by combining the medical and technical terms using Boolean operator such as “AND” and “OR”. An example of the search query used is as follows: (“temporal bone” OR “ear” OR “inner ear” OR “middle ear” OR “ossicles” OR “malleus” OR “incus” OR “stapes” OR “cochlea” OR “facial nerve”) AND (“medical image segmentation” OR “deep learning” OR “CNN” OR “transformer” OR “U-Net” OR “attention mechanism” OR “Dice similarity coefficient” OR “multi-scale feature extraction” OR “small object segmentation”). This strategy provides an exhaustive and technically grounded framework for identifying relevant studies on otologic structure segmentation. The literature search was conducted independently by the investigators. Any discrepancies were resolved through active discussion and consensus.

2.2 Inclusion and exclusion criteria

Studies were included in this review if they met the following criteria:

- Peer-reviewed research articles published in the English language between 11 February 2020 and 19 March 2025.
- Studies focusing on the segmentation of one or more otologic anatomical structures, including structures of the outer, middle, or inner ear.
- Studies using DL-based segmentation models, such as CNNs, transformer architectures, diffusion mechanisms, graph neural networks, or hybrid approaches.
- The studies address the specific challenges associated with segmenting otologic structures and employing relevant quantitative evaluation metrics to assess their performance.
- The studies utilizing relevant medical imaging modalities for otologic structures, such as CT or MRI.

Review articles, meta-analyses, editorials, letters to the editor, and conference abstracts were excluded. Although conference literature was screened during the study selection process, no conference papers met the predefined inclusion criteria and were therefore excluded from the final analysis. Studies lacking sufficient methodological details or failing to report relevant quantitative evaluation metrics were also excluded.

2.3 Literature search and study selection

Our structured literature search yielded a total of 1615 records. After removing 460 duplicate entries, 1155 unique articles remained for screening. These were evaluated based on titles and abstracts, and 1098 were excluded for being irrelevant to the scope. The remaining 57 studies were assessed in full, applying predefined inclusion and exclusion criteria. Ultimately, 38 studies were excluded for reasons such as not being DL-based models ($n=21$), not addressing segmentation tasks ($n=14$), or providing insufficient methodological or result-related information ($n=3$), such as missing key experimental details, incomplete performance metrics, or inadequate dataset descriptions, which prevented meaningful comparison. A total of 19 studies met all eligibility criteria and were included in this review. The study searching process was illustrated in Fig. 2 using a PRISMA flow diagram.

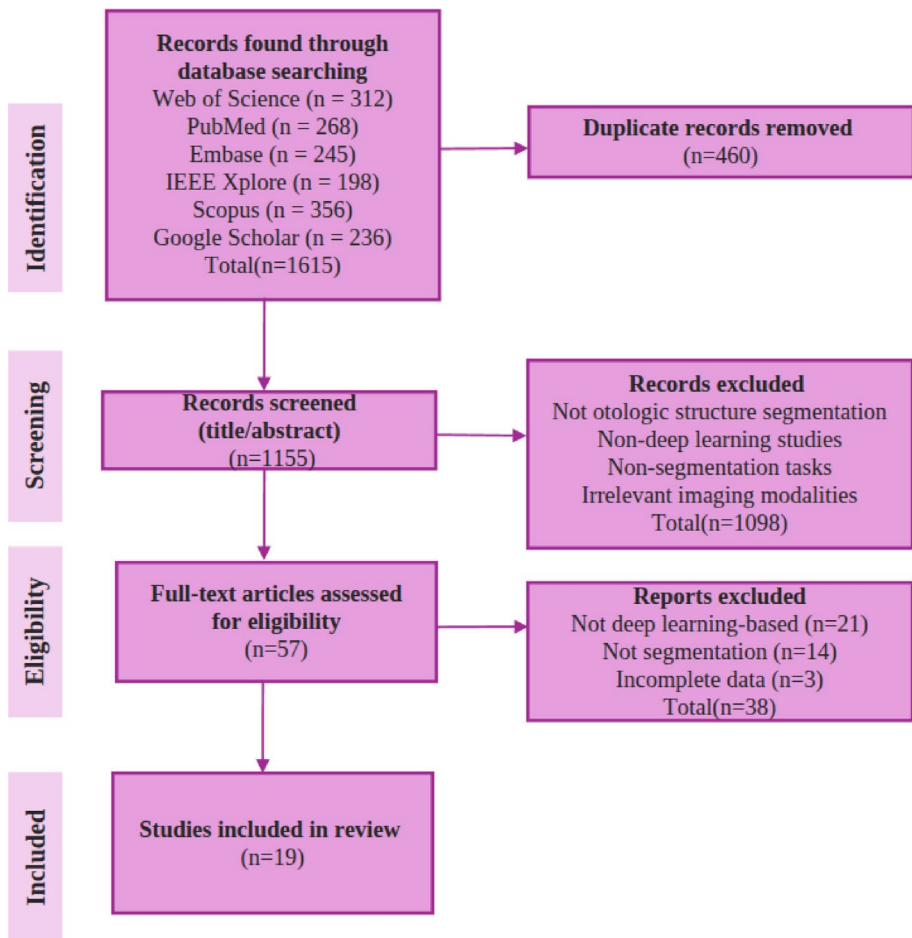


Fig. 2 A PRISMA flowchart illustrating the study selection process for DL-based otologic structure segmentation research

2.4 Quality assessment

Study reporting quality and methodological completeness were evaluated using the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) framework. CLAIM provides a structured set of 39 reporting items designed to assess methodological clarity, transparency, and reproducibility in AI-based medical imaging studies. Each included study was reviewed against all CLAIM checklist items, which cover major reporting domains including title and abstract, study design, dataset description, ground-truth annotation procedures, model architecture and training details, evaluation methodology, and discussion of study limitations (Kim et al. 2023; Zhong et al. 2022; Si et al. 2022; Tejani et al. 2023). For each checklist item, adherence was recorded as present or absent based on information reported in the original articles. The item-level adherence results are summarized in Table 1, which presents both the number and percentage of studies fulfilling each item.

3 Deep learning-based approaches for automated otologic structure segmentation: advances and innovations

The standard steps involved in a DL-based segmentation pipeline are illustrated in this section. The next core step is model training, which involves selecting appropriate architecture, defining loss functions, and optimizing hyper-parameters. Finally, post-processing is performed to refine model outputs, enhance their clinical reliability, and improve spatial coherence. This refinement may include applying morphological operations, contour smoothing, noise removal, and conditional random fields.

3.1 Quality assessment results overview of deep learning architectures for otologic structure segmentation

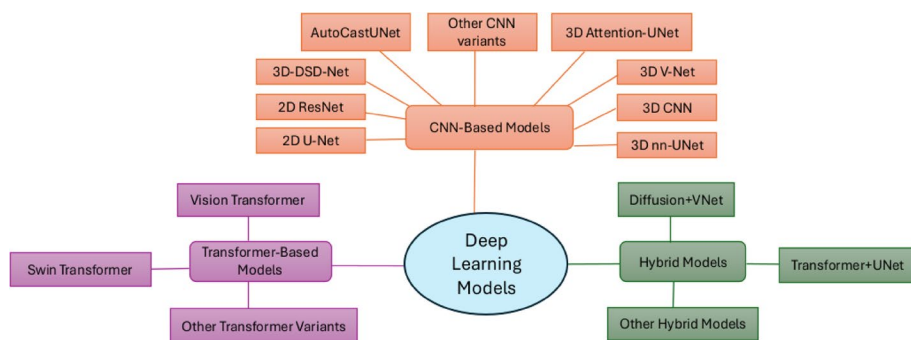
The study reporting quality assessment with the CLAIM framework revealed substantial variability in reporting completeness across the included studies (Table 1). Core reporting elements such as identification of AI methodology, study objectives, dataset sources, and performance metrics were consistently described, with the nine checklist items showing full adherence (100%). However, several critical methodological aspects were documented less consistently. In the data reporting domain, dataset sources were described in all studies (100%), and eligibility criteria (74%) and data preprocessing procedures (84%) showed relatively high adherence. In contrast, de-identification methods (16%) and handling of missing data (21%) were rarely described. Similarly, reporting related to annotation quality was inconsistent; while annotation sources were described in 74% of studies, inter- and intra-rater variability assessments were reported in only 32%. Model development and training details were generally well described, with all studies providing a detailed model architecture description (100%) and a substantial portion of studies reporting training strategies (74%). However, reporting of model initialization procedures (47%) and ensemble techniques (11%) was less frequent. Evaluation practices also varied across studies; while performance metrics were reported in all studies (100%), statistical uncertainty measures such as confidence intervals (26%), robustness analyses (21%), and explainability methods (16%) were rarely described. External validation was reported in only 32% of studies,

Table 1 CLAIM-based reporting quality assessment

Section and topic	Item #	Checklist item	Number of articles adhered
Title or Abstract	1	Identification as a study of AI methodology, specifying the DL technology used	19/19 (100%)
	2	Structured summary of study design, methods, results, and conclusions	19/19 (100%)
Introduction	3	Scientific and clinical background, including the intended use and clinical role of the AI approach	19/19 (100%)
	4	Study objectives and hypotheses	19/19 (100%)
Methods	5	Prospective or retrospective study	17/19 (89%)
	6	Study goal, such as model creation, exploratory study, feasibility study, non-inferiority trial	19/19 (100%)
Data	7	Data sources	19/19 (100%)
	8	Eligibility criteria: how, where, and when potentially eligible participants or studies were identified (e.g., symptoms, results from previous tests, inclusion in registry, patient-care setting, location, dates)	14/19 (74%)
	9	Data pre-processing steps	16/19 (84%)
	10	Selection of data subsets, if applicable	17/19 (89%)
	11	Definitions of data elements, with references to Common Data Elements	8/19 (42%)
	12	De-identification methods	3/19 (16%)
	13	How missing data were handled	4/19 (21%)
	14	Definition of ground truth reference standard, in sufficient detail to allow replication	18/19 (95%)
Ground Truth	15	Rationale for choosing the reference standard (if alternatives exist)	12/19 (63%)
	16	Source of ground-truth annotations; qualifications and preparation of annotators	14/19 (74%)
	17	Annotation tools	7/19 (37%)
	18	Measurement of inter- and intra-rater variability; methods to mitigate variability and/or resolve discrepancies	6/19 (32%)
Data partitions	19	Intended sample size and how it was determined	5/19 (26%)
	20	How data were assigned to partitions; specify proportions	17/19 (89%)
	21	Level at which partitions are disjoint (e.g., image, study, patient, institution)	12/19 (63%)
	22	Detailed description of model, including inputs, outputs, all intermediate layers and connections	19/19 (100%)
	23	Software libraries, frameworks, and packages	15/19 (79%)
	24	Initialization of model parameters (e.g., randomization, transfer learning)	9/19 (47%)
	25	Details of training approach, including data augmentation, hyper-parameters, number of models trained	14/19 (74%)
	26	Method of selecting the final model	11/19 (58%)
Evaluation	27	Ensemble techniques, if applicable	2/19 (11%)
	28	Metrics of model performance	19/19 (100%)

Table 1 (continued)

Section and topic	Item #	Checklist item	Number of articles adhered
Results	29	Statistical measures of significance and uncertainty (e.g., confidence intervals)	5/19 (26%)
	30	Robustness or sensitivity analysis	4/19 (21%)
	31	Methods for explain-ability or interpret-ability (e.g., saliency maps), and how they were validated	3/19 (16%)
	32	Validation or testing on external data	6/19 (32%)
	33	Flow of participants or cases, using a diagram to indicate inclusion and exclusion	10/19 (53%)
Data	34	Demographic and clinical characteristics of cases in each partition	11/19 (58%)
Model performance	35	Performance metrics for optimal model(s) on all data partitions	19/19 (100%)
	36	Estimates of diagnostic accuracy and precision (such as 95% confidence intervals)	14/19 (74%)
	37	Failure analysis of incorrectly classified cases	7/19 (37%)
Discussion	38	Study limitations, including potential bias, statistical uncertainty, and generalizability	16/19 (84%)
	39	Implications for practice, including the intended use and/or clinical role	15/19 (79%)

**Fig. 3** Taxonomy of DL models for otologic structure segmentation

which indicates limited evaluation of model generalizability. Overall, the CLAIM-based quality assessment highlights several persistent reporting deficiencies in the current literature, particularly regarding data handling, external validation, and methodological transparency, which may affect the reproducibility and clinical translation of deep learning-based otologic segmentation models.

To further contextualize these findings, we next examined the architectural trends and evaluation strategies of the included deep learning models for automated otologic structure segmentation. Figure 3 presents the hierarchical categorization of DL models from 2020 to 2025, as reviewed in this study for automated otologic structure segmentation. Table 2 summarizes the key characteristics of these models, including their target anatomical structures,

Table 2 An overview of DL models for automated otologic structure segmentation from 2020 to 2025

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
CochleaSegNet (Heutink et al. 2020) code: *	2020	4: cochlea (scala vestibuli, scala media, scala tympani, helicotrema)	CT, 512×512 , 123 patients	DSC: 0.90 BF: 0.95 mHD: 3.05 aHD: 0.32	2D; CNN-based; Residual CNN	It integrates multiscale residual CNNs, which extract features at multiple spatial resolutions to capture both fine-grained and coarse contextual information; an encoder-decoder architecture, which supports hierarchical representation learning by progressively abstracting features in the encoder and reconstructing spatial detail in the decoder; and morphological thinning, which iteratively removes peripheral pixels from segmentation masks to produce single-pixel-wide boundaries, thereby improving edge precision all within a unified two-stage pipeline.
3DDeep Supervised Densely Network (3D-DSD-Net) (Li et al. 2020a) code: *	2020	9: malleus, incus, external contour of cochlear capsule, internal cavity of cochlear capsule, vestibule, superior semi-circular canal, posterior semi-circular canal, lateral semi-circular canal, internal acoustic meatus	CT, 512×512 , 64 patients	DSC: 0.77 ASD: 0.20 aHD: 0.43	3D; CNN-based; densely connected CNN	It integrates densely connected convolutional blocks, where each layer receives input from all preceding layers within the block to promote efficient feature reuse and strengthen feature propagation. A 3D multi-pooling module is employed to capture multi-scale contextual information by aggregating features through multiple pooling operations with different kernel sizes and strides. In the decoder, deep supervision is applied by attaching auxiliary segmentation outputs at intermediate stages, which enhances gradient flow during training and facilitates more accurate final segmentation.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
PWD-3D Net (Nikan et al. 2020) code: https://github.com/AuditoryBiophysics-Lab/Slicer-ABLTemporalBoneSegmentation	2020	8: sigmoid sinus, facial nerve, inner ear, malleus, incus, stapes, internal carotid artery, internal auditory canal	CT, 144 × 144, 156	DSC: 0.86 HD: 0.755 JS: 0.716	3D; CNN-based; DenseVNet	It integrates balanced window sampling, which selects training patches with a proportional representation of each class to mitigate class imbalance; dense connectivity, which links each layer to all subsequent layers to promote efficient feature reuse and gradient flow; parametric ReLU activation, which adaptively learns the slope of negative responses to improve model expressiveness; and B-spline upsampling, which reconstructs high-resolution outputs using smooth interpolation to minimize checkerboard artifacts. It also adopts a V-shaped encoder-decoder with skip connections, enabling the network to preserve spatial details while enhancing feature retention across scales.
W-Net (Lv et al. 2021) code: https://github.com/DnkiAutomaticSegmentation-of-Temporal-BoneStructures-from-ClinicalConventional-CT	2021	3: cochlear labyrinth, ossicular chain, facial nerve	CT, 512 × 512, 39	DSC: 0.84 ASSD: 0.086 SDD: 0.196	3D; CNN-based; U-Net	It incorporates a shallow W-shaped encoder-decoder architecture designed to preserve spatial resolution throughout the feature extraction and reconstruction process. To ensure the retention of fine-grained details, three skip connections are employed, enabling the transfer of high-resolution features directly to the decoding path. Furthermore, these skip connections are intentionally decoupled from the overall network depth, preventing the progressive disappearance of small targets in deeper layers. These design choices enhance the model's ability to accurately delineate small-scale anatomical structures by maintaining spatial precision and feature integrity across the network.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation type, architecture, and backbone	Core technological innovation
AH-Net (Neves et al. 2021) code: *	2021	4: inner ear, ossicles, facial nerve, sigmoid sinus	CT, 256 × 256, 150	DSC: 0.83 TS: 4.72 s HD: 0.33 VS: 102.38	2D/3D; CNN-based hybrid; U-Net + ResNet	It integrates 3D U-Net-based volumetric context modeling to capture spatial relationships in images, anisotropic 3D convolutions in the encoder to reduce computational cost, a 2D ResNet-based decoder enhances inference efficiency and generalization, and a 3D-to-2D hybrid design to enable fast, memory-efficient, and accurate segmentation.
AutoCasU-Net (Hussain et al. 2021) code: https://github.com/raabid236/AutoCasNet .	2021	3: cochlear scalae, semicircular canals, vestibule	CT, 256 × 256, 17	DSC: 0.90 HD: 0.74	2D; CNN-based cascaded architecture; Cascaded U-Net	It integrates a three-stage cascaded CNN for sequential refinement across axial, coronal, and sagittal views; 2D U-Net-based segmentation at each stage to independently process slices; 3D volume reconstruction with largest connected component extraction to enforce anatomical plausibility; and a hybrid design that progressively incorporates spatial context to balance volumetric consistency with computational efficiency.
W-Net (Wang et al. 2021b) code: *	2021	3: facial nerve, cochlear labyrinth, ossicles	CT, 512 × 512, 58	DC (normal): 0.82 DSC (abnormal): 0.66 ASSD (normal): 0.14 ASSD (abnormal): 0.91	3D; CNN-based; W-Net	It incorporates a dual-path encoder-decoder architecture, where one path focuses on capturing global contextual features while the other preserves fine local details, enabling complementary feature representations for enhanced reconstruction. It employs 3D convolutions with zero-padding to maintain the original spatial dimensions after convolution, thereby preserving anatomical integrity across layers. Additionally, a gradual channel-reduction strategy is used in the decoder to progressively compress feature maps while retaining high-resolution anatomical details.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
3D U-Net (Vaidyanathan et al. 2021) code: *	2021	4: inner ear, cochlear capsule, vestibule, semi-circular canals	MRI, $256 \times 256, 177$	DSC: 0.87 TPR: 0.915 FNR: 0.0849 FDR: 0.148	3D; CNN-based; 3D U-Net with attention gates	It extends the classical 3D U-Net by integrating attention gates within skip connections to selectively propagate relevant features; incorporating a multi-scale input strategy using an image pyramid to capture contextual information at multiple resolutions; deepening the encoder by expanding convolutional filters from 16 to 128 to enhance feature representation capacity; replacing batch normalization with instance normalization to improve robustness across scans with varying intensity distributions; and employing transposed 3D convolutions in the decoder to enable efficient spatial up sampling for accurate volumetric segmentation.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
3D V-Net (Wang et al. 2022b) code: *	2022	3: malleus, incus, stapes	CT, 1024×1024 , 158 patients	DSC: 0.89 ASD: 0.25 HD95: 1.04	3D; CNN-based; V-Net (VB-Net variant)	It introduces VB-Net, based on V-Net, by integrating compact residual blocks to reduce the number of trainable parameters while enabling deeper representation learning; adopts a cascaded dual-resolution framework to first localize approximate anatomical structures at low resolution and then refine boundary details at high resolution; employs a resolution-aware coarse-to-fine segmentation strategy to capture global context while preserving fine details; incorporates a shallow W-shaped encoder-decoder architecture to maintain spatial resolution during feature extraction and reconstruction; and utilizes three strategically designed skip connections, decoupled from overall network depth, to transfer high-resolution features directly into the decoding path and prevent the disappearance of small targets in deeper layers, thereby enhancing the model's ability to delineate delicate anatomical structures with spatial precision and feature integrity.
W-Net (Ke et al. 2023) code: *	2023	7: facial nerve, inner ear, lateral semi-circular canal, superior and posterior semi-circular canals, internal auditory canal, internal carotid artery, glomus jugulare	CT, 512×512 , 80	DSC: 0.80 ASSD: 0.21	3D; CNN-based; Modified W-Net	It integrates a modified 3D CNN architecture with dual encoder-decoder paths to enhance multi-scale representation, cross skip connections to improve feature interaction, high-resolution input feature maps to enable precise localization, and parameter sharing across decoder branches to reduce memory usage and accelerate convergence, resulting in a lightweight network with less than one-tenth the parameters of a standard 3D U-Net.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation type, architecture, and backbone	Core technological innovation
no new U-Net (nnU-Net) (Isensee et al. 2021) code: *	2023	7: ossicles, inner ear, facial nerve, chorda tympani, bony cochlear labyrinth, internal auditory canal, internal carotid artery	CT, 512 × 512, 15 patients	DSC: 0.84 mHD: 0.084	3D; CNN-based self-configuring architecture; nnU-Net	It leverages a self-configuring architecture that automatically adapts preprocessing and training to the specific medical dataset. This includes resampling and intensity normalization to standardize inputs, dynamic adjustment of network depth, kernel size, and patch dimensions to match anatomical resolution, tailored data augmentation to enhance generalization, rigid co-registration to ensure geometric consistency, patch-based training with cross-entropy and dice loss to address class imbalance, and sliding-window inference with Gaussian-weighted stitching to generate accurate, artifact-free volumetric segmentations. This fully automated pipeline enables scalable and robust segmentation of small, morphologically diverse structures without manual hyper-parameter adjustment.
3D U-Net (Li et al. 2023b) code: *	2023	3: cochlear anatomy (bony cochlea, internal cochlear structures, membranous cochlea)	CT, 384 × 384, 100	DSC: 0.74 HD95: 4.217	3D; CNN-based; Residual 3D U-Net	It extends the standard 3D U-Net architecture by integrating residual blocks within the encoder path to improve feature reuse, incorporating dropout regularization to prevent over-fitting, applying instance normalization to enhance convergence and generalization, and adopting LeakyReLU activation to ensure gradient flow in low-contrast regions. It also employs a multi-scale feature fusion strategy to capture both global context and fine anatomical details and utilizes patch-wise training with skip connections to preserve high-resolution information while reducing computational cost.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
Res U-Net (Zhou and Li 2023) code: *	2023	4: Ossicular chain, internal auditory canal, facial nerve, cochlear labyrinth	CT, 512 × 512, 147	DSC: 0.82 HD95: 0.6139	3D; CNN-based residual architecture; Res U-Net	It integrates residual units into both encoder and decoder paths of the U-Net, replacing traditional concatenation-based skip connections with additive residual mappings. This design improves feature reuse, gradient flow, and structural consistency across layers. Additionally, it repositions the upsampling and downsampling operations to the start of each block, allowing more effective spatial feature transformation while maintaining matched volume dimensions between encoder and decoder stages.
Multi-Task Dual-Decoder 3D U-Net (Stebani et al. 2023) code: https://github.com/stebix/cochlea-spatial-investigation	2023	4: cochlear capsule; cochlear windows (oval window, round window); helicotrema	CT, 128 × 128 × 128, 52	DSC: 0.95 IOU: 0.91 aHD: 0.102	3D; CNN-based multi-task architecture; Dual-decoder 3D U-Net	It introduces a multi-task 3D U-Net featuring a shared encoder and two dedicated decoders: one for semantic segmentation and the other for anatomical landmark localization. To promote spatial consistency and information flow across tasks, it employs cross-decoder skip inter-connections, wherein intermediate feature maps from the segmentation pathway are reused in the heatmap regression branch. This enables precise localization by enriching the landmark decoder with fine-grained spatial details from the segmentation decoder. Each decoder maintains separate weights to allow task-specific feature refinement. Additionally, a composite loss function and landmark heatmap accuracy, enabling simultaneous structure delineation and coordinating prediction from a single volumetric input.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
U-Net CNN (Lim et al. 2024) code: https://github.com/perslev/	2024	3: inner ear (cochlea, vestibule, semicircular canals)	CT, not mentioned, 271 patients	DSC: 0.83 LS: 1	3D; CNN-based patch-wise architecture; U-Net	It adapts open-source U-Net framework into a patch-based segmentation pipeline to focus computational resources on the most relevant anatomical regions, employs a symmetrical encoder-decoder architecture to extract high-level semantic features in the contracting path and progressively restore spatial resolution in the expanding path, incorporates skip connections to ensure fine detail recovery during up sampling and mitigate the loss of small structural information during down sampling.
UNETR (Li et al. 2023b) code: *	2023	4: Cochlea (scala vestibuli, scala media, scala tympani); helicotrema	CT, 64 × 64, 92	DSC: 0.92	3D; Transformer-based former-based hybrid; ViT encoder+U-Net decoder	It integrates a transformer-based encoder into the classic U-Net architecture to capture global multiscale context and long-range dependencies, processes volumetric data as sequences of patch embeddings with multi-headed self-attention (MHSA) and multi-layer-perceptron (MLP) blocks to enhance contextual understanding and connect directly to the decoder via multi-resolution skip connections to enable precise spatial reconstruction.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
SwinUNETR (Neves et al. 2024) code: *	2024	8: internal carotid artery, chorda tympani, external auditory canal, internal auditory canal, facial nerve, inner ear, ossicles, ossicular chain	CT, 128 × 128, 325 patients	DSC: 0.87 aHD: 0.79 aPT: 9.1 s	3D; Hybrid CNN–Transformer (patch encoder); Swin Transformer encoder + U-Net decoder	It utilizes a hierarchical Swin Transformer encoder that partitions a volume into non-overlapping 3D patches, applies shifted window-based self-attention within localized regions to capture long-range spatial dependencies, and progressively merges patches across stages to generate multi-scale feature representations. It further integrates a U-Net-based decoder that receives the hierarchical features via skip connections, employs transposed convolutions and convolutional refinement blocks to progressively up sample feature maps, and reconstructs anatomically precise segmentation outputs by fusing local and global contextual information.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture, and backbone	Core technological innovation
Standard observation plane (SOP) Graph model (Li et al. 2025a) code: *	2025	3: ossicular chain (malleus, incus, stapes)	CT, 650 × 650 × 370, 640 patients	Time: the annotation time of one case was reduced from 12 min to about 2 min	2D; Graph-guided hybrid; TransUNet + SOP Graph	It addresses the challenge of expert-dependent, time-intensive SOP annotation in otologic CT imaging by transforming radiologist interpretation logic into a graph-based framework. It introduces a directed SOP-Graph where each node represents a parameterized observation plane and edges encode spatial dependencies and diagnostic localization sequences, enabling structured representation of expert knowledge. It employs a heuristic SOP localization algorithm that operates on segmentation-derived ossicular masks, classifies SOPs based on localization complexity, and applies slice-based search with axis-specific rotation estimation to detect standard and non-standard observation planes. It further integrates a graph-matching verification step using a radiologist-curated SOP reference set to automatically identify and filter failed annotations. This framework significantly reduces annotation time while preserving spatial consistency, anatomical accuracy, and diagnostic interpretability: spatial consistency is ensured by modeling SOPs as a connected graph with defined relative orientations; anatomical accuracy is preserved by anchoring plane localization on ossicular segmentations; and diagnostic interpretability is maintained through radiologist-defined SOP criteria and verification against expert-curated templates.

Table 2 (continued)

DL models	Year	Target structure(s)	Clinical dataset attributes (Modality, image resolution, number of patients)	Evaluation metrics	Input representation, architecture type, and backbone	Core technological innovation
Semi-supervised ear bone segmentation (SemiEMBS) (Xie et al. 2025) code: https://github.com/xmed-lab/SemiEBS	2025	6: ossicular chain (malleus, incus, stapes); bony cochlear labyrinth (scala vestibuli, scala tympani, scala media)	CT, not described, 304 patients	DSC: 0.94	3D; Semi-supervised diffusion-based hybrid; Diffusion encoder + V-Net decoders	It introduces a semi-supervised otology segmentation model that employs a diffusion-based V-Net encoder to aggregate domain-invariant features from both labeled and unlabeled HRCT data, integrates dual decoders trained on labeled inputs to preserve segmentation quality and generate high-fidelity pseudo labels, and incorporates an additional V-Net decoder trained exclusively on unlabeled inputs to mitigate overfitting and enhance generalization. It further enables automated 3D segmentation and reconstruction of otologic structures such as the ossicular chain and bony labyrinth by leveraging pseudo-label supervision and diffusion-driven feature representation to accurately capture small, irregular ear bones, thereby achieving high dice accuracy while significantly reducing manual annotation burden and processing time and preserving anatomical precision across diverse otologic pathologies.

*Indicate that there is no publicly available code for the DL models

clinical datasets, image dimensionality, backbone architecture, evaluation metrics, and core technological innovations. For model training, the input data may be provided either in two-dimensional (2D) form such as individual CT or MRI slices or as full 3D volumes composed of sequential slices that preserve spatial continuity. This distinction is important because 2D-based models operate on a per-slice basis, whereas 3D-based models may use spatial anatomical relationships across the entire volume. The choice of evaluation metrics is equally important, as they provide objective measures of segmentation performance. In otologic structure segmentation, performance evaluation requires a comprehensive set of metrics that capture different aspects of accuracy, boundary precision, volumetric fidelity, and computational efficiency. Overlap-based metrics such as Dice similarity coefficient (DSC), intersection over union (IoU), and Jaccard score (JS) are commonly reported. Boundary and distance-based metrics include Hausdorff distance (HD), HD-95, average Hausdorff distance (aHD), maximum HD (mHD), average surface distance (ASD), average symmetric surface distance (ASSD), and boundary F-measure (BF). Volumetric agreement is captured by volumetric similarity (VS) and lesion size (LS) measures. Classification performance is described using true positive rate (TPR/sensitivity), false negative rate (FNR), false discovery rate (FDR), and specificity. Practical feasibility is assessed via time per segmentation (TS) and average processing time (aPT).

Together, these metrics form a comprehensive evaluation framework. Overlap-based scores quantify global accuracy, distance-based measures capture local boundary precision, volumetric metrics ensure structural fidelity, classification rates evaluate detection reliability, and efficiency measures assess clinical practicality. By combining these perspectives, segmentation performance can be evaluated not only in terms of accuracy but also robustness, interpretability, and readiness for real-world deployment in otologic imaging. To improve readability, the architecture column explicitly reports the input dimensionality, architecture category, and backbone design for each method, distinguishing CNN-based, transformer-based, hybrid CNN–Transformer, graph-guided, and diffusion-based approaches.

3.2 Convolutional neural network-based approaches

CNNs have become a widely adopted architecture for medical image segmentation due to their ability to automatically learn spatial and hierarchical features from imaging data (Ali et al. 2021; Masood et al. 2024; Müller and Kramer 2021). Prior studies have demonstrated the effectiveness of CNN-based methods in segmenting various anatomical structures, including brain tissue (Mostafa et al. 2023), neuroanatomical regions (Swathi and Sankaran 2025), the prostate (Wang et al. 2024a) (Gade et al. 2024), heart (Singh et al. 2025; Cuellar et al. 2024; Li et al. 2020b), cranial and appendicular bones (Wang et al. 2024b; Boutillon et al. 2022) and tumors (Sahoo et al. 2023; Kashyap et al. 2025) often surpassing traditional image processing techniques (Archana and Jeevaraj 2024). In the context of otologic structures, CNNs facilitate accurate segmentation of small and intricate anatomical components such as the stapes, incus, and facial nerve structures that are particularly challenging due to their small size, low contrast against surrounding bone, and morphological variability. Figure 4 illustrates a standard CNN architecture applied to otologic CT imaging data.

Overall, CNN-based architectures remain important and effective foundations for otologic structure segmentation because they preserve local spatial detail, support efficient hierarchical feature extraction, and have demonstrated strong performance across several

temporal bone segmentation tasks. Their limitations should therefore be interpreted in context rather than as fundamental weaknesses. Challenges related to limited receptive field, computational cost in 3D implementations, and dependence on expert annotations have motivated the development of hybrid CNN–Transformer, attention-based, and semi-supervised approaches. These newer models are best viewed as complementary extensions that address specific limitations while retaining the local feature-learning strengths of CNNs.

3.2.1 U-net and its variants

Among CNN-based architectures, U-Net (Ronneberger et al. 2015) remains the most widely used CNN architecture for otologic segmentation due to its encoder–decoder structure with skip connections that preserve spatial detail while integrating deep semantic context. These properties make U-Net effective for segmenting very small structures such as the incus, stapes and malleus (Tsuno et al. 2022). However, the vanilla design suffers from loss of fine details during repeated down sampling and semantic mismatch between encoder and decoder features, which reduces precision in low-contrast regions.

To address these limitations, Johannes Fauser et al. (Fauser et al. 2019) introduced a U-Net variant incorporating shape-regularization and trajectory planning, supported by probabilistic active shape models to reduce the semantic gap. Their 2D slice-based approach was computationally efficient but lacked full 3D spatial continuity. Jannik Stebani et al. (Stebani et al. 2023) overcame this by proposing a 3D attention U-Net that processed complete CT volumes and used attention gates to filter irrelevant encoder features before fusion, achieving high accuracy (average DSC $\approx 95\%$) and reduced false positives for otologic structures.

3.2.2 ResNet-based architectures

ResNet architectures improve otologic segmentation through residual blocks, which preserve fine-grained spatial cues and stabilize gradient flow in deep networks, critical advantages for modeling small, low contrast otologic anatomy. Caio Neves et al. (Neves et al. 2021) applied ResNet-101 to segment four otologic structures in 150 CT scans, achieving an average DSC of 0.83. Langtao Zhou et al. (Zhou and Li 2023) proposed SegResNet and ResU-Net, both using residual learning to enhance feature retention. SegResNet employs a deep encoder and lightweight decoder with residual blocks to support hierarchical representation learning while managing computational cost, whereas ResU-Net embeds residual blocks throughout both encoder and decoder and adds local skip connections for better preservation of fine anatomical boundaries.

By improving feature preservation, reducing the semantic gap between encoding and decoding, and stabilizing learning in deep networks, U-Net enhancements and ResNet-based models significantly improve boundary precision for tiny otologic structures. These developments reduce false positives and false negatives in diagnostically sensitive regions and enable more reliable segmentation across diverse datasets, supporting future applications in image-guided surgery, hearing-health care planning, and automated anatomical assessment.

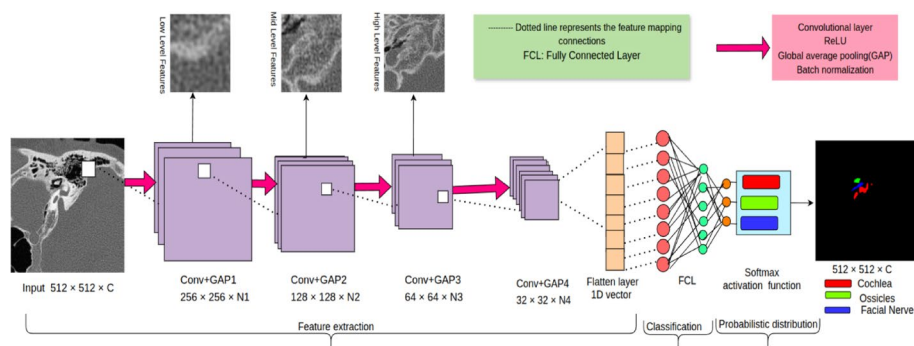


Fig. 4 Illustration of a CNN architecture

3.2.3 Other CNN-based architectures for otologic structure segmentation

Beyond U-Net and ResNet families, several CNN architectures have been developed to address other challenges such as redundant feature handling, class imbalance, limited global context, and computational inefficiency.

Xiaoguang Li et al. (2020a) proposed a 3D deeply supervised, densely connected encoder–decoder network to improve segmentation of small otologic structures. In this architecture, dense connectivity was implemented such that each layer in the encoder and decoder received feature maps from all preceding layers in the same block, ensuring maximal feature reuse, stronger gradient flow, and thus preserving fine spatial details. This connectivity, combined with multi-scale fusion during down sampling, allowed the network to retain fine-grained cues while capturing broader contextual information essential for accurately identifying tiny, low contrast otologic structures embedded within complex bony surroundings. When evaluated on a dataset of 64 CT patient studies across nine anatomical structures, the model achieved DSC scores ranging from 68.02% to 83.62%, with major structures exceeding 82%. However, the 3D dense connectivity and fusion strategy resulted in high computational burden, increased memory usage, and prolonged training time. Moreover, the presence of redundant feature propagation where multiple feature maps contributed overlapping information limited efficiency gains. These limitations encouraged later studies to explore more computationally efficient architectures, including optimized 2D and hybrid designs, while preserving the strengths of CNNs in local feature extraction.

Jia Ke et al. (2023) addressed the computational inefficiencies and redundancy issues present in prior 3D architectures by proposing a W-shaped dual encoder–decoder network that utilizes 2D convolutions and optimized cross skip connections for efficient multi-scale feature reuse. By processing features in parallel across multiple spatial scales and merging them through strategically designed fusion paths, the architecture enhanced feature representation while significantly reducing parameter count and memory consumption. Parameter sharing between decoder branches further streamlined the model, enabling faster convergence and efficient training. Evaluated on a dataset of 80 CT patient studies from both adult and pediatric cohorts, the model demonstrated improved segmentation performance in terms of both accuracy and computational efficiency across eleven otologic structures, outperforming Xiaoguang Li et al.’s 3D dense model (Li et al. 2020a). However, despite these advantages, the use of fixed convolutional configurations and conventional loss func-

tions constrained its ability to capture the most intricate anatomical boundaries, especially for extremely small or low-contrast structures, suggesting potential for further refinement.

Built on Xiaoguang Li et al. (2020a), a work proposed by Jiang Wang et al. (2021a) targeted two persistent challenges: first, inflating memory and computation due to the redundant feature handling. Jiang Wang et al. improved feature propagation efficiency by proposing a refined W-Net variant with 3D convolutions using “same” (pixel) padding and optimized channel-depth transitions. Secondly, this work further addresses the issue of class imbalance, a common challenge in otologic datasets where target ROI are vastly outnumbered by background voxels. To mitigate this, a weighted cross-entropy loss was employed, assigning greater importance to errors on small structures to ensure they are not overshadowed during the optimization process. The model was evaluated on a dataset of 39 CT patient studies covering three otologic structures and achieved better DSCs than Xiaoguang Li et al.’s model (2020a). Nonetheless, its increased depth added computational overhead, and repeated down sampling continued to risk loss of fine, low-contrast boundaries.

To reduce the computational cost and detail loss of deep architectures, Yi Lv et al. (2021) designed a shallow encoder–decoder with only a single down sampling stage, keeping features close to native resolution. To retain contextual information despite the reduced depth, the model incorporated three parallel same-resolution branches with varying receptive fields (achieved through different kernel sizes) and fused them via skip connections at the same resolution. This strategy preserved fine details while capturing multi-scale spatial cues. Additionally, shared convolutional weights were employed to reduce parameter count and enforced feature consistency, though at the expense of flexibility in modeling diverse feature maps. The model was evaluated on 30 CT studies, and preserved spatial detail well, but its shallow depth limited hierarchical abstraction and global context modeling capabilities often needed to separate small, low-contrast targets from visually similar surroundings.

To better capture subtle and low-contrast anatomical structures while avoiding the prohibitive computational cost of processing full 3D volumes, Soodeh Nikan et al. (2020) introduced a patch-wise (window-based) analysis strategy. By restricting training and inference to local 3D patches, the network focused more effectively on fine-scale structures, improving sensitivity to small anatomical regions. Dense connectivity within the patch-level architecture further enhanced feature reuse and stabilized gradient flow, leading to richer representations despite the shallow network depth. The resulting PWD-3DNet achieved improved segmentation accuracy across 41 CT patient scans, particularly for localized and low-contrast structures. However, patch-wise inference may reduce access to full-volume global context unless combined with appropriate overlap, aggregation, or post-processing strategies. To reconstruct predictions at the full-volume level, dense sliding-window inference is required an approach that increases computational burden and raises challenges in ensuring consistent segmentation across patch boundaries.

To combine detailed segmentation with efficient computation, Xing-Rui Wang et al. (2022a) introduced a coarse-to-fine cascaded framework. In the first stage, a down sampled full-volume CT was used to rapidly localize the ROIs. In the second stage, only these localized ROIs were cropped and re-analyzed at full resolution to refine boundary accuracy. This two-step strategy reduced memory consumption, balanced the foreground–background voxel ratio within each cropped region, and recovered fine anatomical details often lost in single-step segmentation.

Building on this principle, a cascaded version of VB-Net (a compact 3D V-Net backbone with residual blocks) demonstrated that such high-resolution refinement could be achieved efficiently without the prohibitive cost of full-volume, full-resolution processing. Thus, while Xing-Rui Wang et al. established the general coarse-to-fine cascaded paradigm, VB-Net provided an optimized backbone to implement it with fewer parameters and lower computational overhead. Tested on 158 CT patient studies, both methods achieved high DSC, showing the benefit of first roughly locating structures and then refining them for accurate segmentation of small, low-contrast, and complex otologic anatomy.

3.3 Emerging advances in deep learning for otologic structure segmentation

Although CNN-based architecture remains highly effective for local feature extraction and fine boundary preservation in otologic structure segmentation, some limitations remain when modeling long-range anatomical dependencies and global contextual relationships. CNNs rely on local receptive fields, which can make it challenging to capture relationships between spatially distant structures. Such context may be useful for differentiating small otologic structures from surrounding bone or soft tissue, particularly for structures such as the ossicles or facial nerve. In addition, many CNN-based methods require carefully annotated datasets for optimal performance, and annotation of small, scattered otologic structures remains time-consuming and expertise-dependent.

To overcome these challenges, researchers have recently introduced several advanced DL frameworks. For instance, hybrid CNN–transformer models combined the local feature extraction strengths of CNNs with the global attention mechanisms of transformers, enhancing contextual understanding and enabling more accurate segmentation of spatially intricate structures.

3.3.1 Transformer-based hybrid architectures

Transformer-based architectures have gained particular attention due to their ability to model global contextual relationships through self-attention mechanisms. Unlike CNNs, which process data in a spatially localized manner, transformers attend to all spatial positions simultaneously, enabling a global relational representation of structures distributed within otologic structures. As illustrated in Fig. 5, the vision transformer architecture comprises several key components. The process begins with the input image, which is divided into fixed-size non-overlapping image patches. These patches are flattened and passed through a linear patch embedding layer, which projects each patch into a high-dimensional vector space. A positional embedding is then added to preserve spatial ordering, with no [class] token included, as the network is designed for voxel-wise segmentation rather than image-level classification. These embedded tokens are fed into a stack of transformer encoder blocks. Each block consists of repeated layers (denoted L_x , where L is the number of layers) comprising multi-head self-attention (MHSA) and a feed-forward MLP, with layer normalization preceding and residual connections following each sublayer.

Transformers use self-attention to compute pairwise interactions between all image pixels, enabling each pixel to attend semantically relevant regions across the image regardless of spatial proximity. This global context modeling enhances feature representation, generating more discriminative and context-aware embeddings essential for tasks such as multi-

organ segmentation (Li et al. 2025a; Salimi et al. 2025) and intricate, local structure (Liu et al. 2025b; Xu et al. 2025; Chen et al. 2025). The core mechanism computes similarity between patches by generating query (Q), key (K), and value (V) matrices via 1×1 convolutions. A scaled dot-product between Q and K, followed by SoftMax normalization, generates an attention map that reweighs V to capture global contextual relationships. This resulting self-attended features are refined with another 1×1 convolution and aggregated by the MLP head for semantic segmentation.

For otologic structure segmentation, an example of a successful hybrid approach combining CNNs and transformers is presented by Caio Neves et al. in SwinUNETR (Neves et al. 2024), which integrates the Swin Transformer and CNN as an encoder with a CNN-based decoder. The Swin Transformer encoder divides the input CT volume into non-overlapping 3D patches, which are embedded and processed through hierarchical self-attention with shifted windows. This design enables the model to capture both local contextual details within each patch and long-distance relationships across patches, preserving global semantic continuity throughout the 3D volume. During inference, patch-level predictions are recombined using a sliding-window strategy with an appropriate overlap and blending, ensuring smooth transitions and anatomically consistent 3D segmentations. This structure is particularly effective for capturing the spatial relationships of small and spatially distant otologic substructures, within the full volume. The CNN decoder reconstructs high-resolution segmentation maps using up sampling and localized convolutions, thereby recovering fine structural details such as narrow boundaries and subtle intensity variations. Skip connections between the encoder and decoder enable multi-scale feature fusion, ensuring that both global context and local precision are preserved. This encoder-decoder configuration effectively combines the global modeling capabilities of transformers with the spatial detail-preservation of CNNs, resulting in anatomically consistent and robust segmentation performance. This model is particularly advantageous in otologic imaging, where the target structures such as the incus, stapes, and facial nerve are extremely small, embedded within dense otologic anatomy, and often separated by substantial spatial distances in 3D space. Its hierarchical self-attention mechanism enables us to capture the long-distance anatomical relationships while preserving the fine structural details needed for sub-millimeter boundary delineation. The model was evaluated on a dataset comprising 325 CT patient studies and achieved an average DSC of 87% for segmenting nine key otologic structures.

Another hybrid model, UNETR, proposed by Zhenhua Li et al. (2023b), is a U-shaped segmentation architecture that employs a purely transformer-based encoder combined with a CNN-based decoder. The encoder used a vision transformer to process the input 3D volume by dividing it into non-overlapping patches, embedding them, and applying multiple transformer blocks to model long-range dependencies and capture semantically rich global context. No convolutional operations are used in the encoding stage, allowing the model to fully exploit the self-attention mechanism for volumetric representation learning. To preserve the spatial detail, simple skip connections transfer feature representations from different depths of the transformer encoder to the corresponding stages of the CNN decoder, enabling effective integration of global semantic context with fine-grained spatial information. The decoder then reconstructs high-resolution segmentation maps through progressive up sampling followed by localized convolutional refinements, ensuring accurate boundary delineation and structural consistency. This design is particularly effective for otologic structure segmentation tasks where modeling complex global relationships while preserving

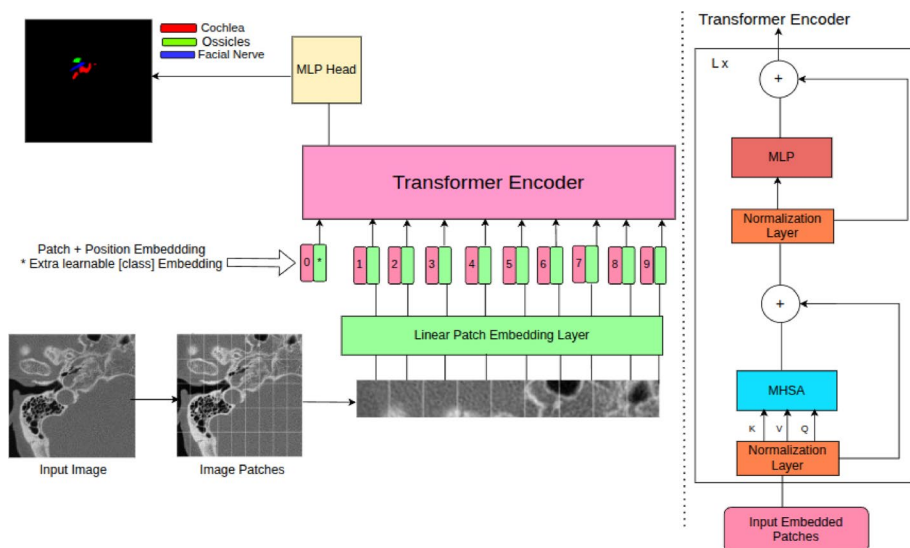


Fig. 5 Overview of the vision transformer architecture for otologic structure segmentation

local anatomical accuracy. The model was evaluated on a dataset comprising 92 CT patient studies for cochlea segmentation, achieving a DSC of 92%.

3.3.2 Diffusion-based hybrid architecture

CNN–diffusion hybrid models integrate convolutional feature extraction with iterative denoising steps, progressively refining segmentation maps by removing high-frequency noise and smoothing irregular boundaries while preserving anatomically valid edges. In contrast to conventional CNNs that primarily emphasize local feature extraction, the diffusion-based encoder captures domain-invariant and fine-grained anatomical representations through a probabilistic denoising process informed by anatomical priors. This guided refinement enables the suppression of spurious responses arising from CT artifacts or granular noise, while preserving the sharp boundary contours of small and structurally complex regions. In low-contrast regions, where subtle intensity variations are easily obscured by noise, this refinement improves boundary precision, ensures structural continuity, and reduces false positives, yielding segmentations that more closely align with true anatomy.

Figure 6 shows a standard forward-backward CNN-diffusion model. The forward process $q(S_T|S_{T-1})$ simulates the gradual degradation of anatomical detail by adding gaussian noise to a clean image S_0 over multiple time steps, culminating in a noise-only image S_T . The reverse process $p_\theta(S_{T-1}|S_T)$, parameterized by a CNN model, learns to invert this corruption by progressively denoising the image and reconstructing high-fidelity anatomical structures.

In otologic structure segmentation, where small anatomical structures and limited high-quality labeled datasets make manual annotation both challenging and time-consuming, semi-supervised learning offers an effective solution by leveraging abundant unlabeled HRCT scans to enhance training efficiency and generalization. A notable approach is the

SemiEBS framework proposed by Mingwei Xie et al. (2025), which introduces the first diffusion-based strategy in this domain to improve learning from limited annotated data.

Semi-supervised training in SemiEBS is achieved through two parallel flows: a labeled flow, where a Diff-VNet encoder–decoder is trained alongside a vanilla V-Net decoder to extract domain-invariant and anatomically relevant features from labeled data; and an unlabeled flow, where high-quality pseudo-labels derived from domain- and class-unbiased predictions are used to supervise a third decoder trained solely on unlabeled data. This dual-path design mitigates overfitting and encourages the network to learn complementary feature representations.

To further strengthen learning from unlabeled inputs, the framework integrates techniques such as consistency regularization and adversarial learning, which ensure stable predictions under perturbations and enforce anatomical plausibility. Additionally, its diffusion-based denoising mechanism guided by structural priors helps recover fine spatial boundaries and reduce noise, enabling accurate reconstruction of delicate ear bone structures such as the ossicular chain and bony labyrinth.

Evaluated on a dataset of 334 HRCT patient studies, SemiEBS achieved an average DSC of 94.76, demonstrating strong generalization across varied otologic pathologies while reducing reliance on manual labels and significantly accelerating 3D reconstruction.

3.3.3 Neural graph-based architecture

Another state-of-the-art approach that addresses critical limitations in otologic structure analysis is the SOP-Graph neural framework. Traditional CNN-based segmentation models process volumetric data voxel-by-voxel or slice-by-slice, requiring the extraction and convolution of high-dimensional feature maps across hundreds of slices. This leads to substantial computational costs during both training and inference. In otologic imaging, where target structures such as the ossicles, semicircular canals, and facial nerve occupy only a minute fraction of the overall image volume, a large portion of computation is wasted on background regions—slowing model convergence and intensifying class imbalance.

To address these limitations, Xiaoguang Li et al. (2025b) proposed the SOP-Graph model, whose primary purpose is to automate and accelerate the annotation of SOPs in HRCT. Manual SOP annotation essential for ossicular chain assessment is typically a time-consuming and expert-dependent task, often taking around 12 min per case due to the small size, oblique orientation, and complex 3D positioning of ossicular structures. The SOP-Graph framework replaces this manual process with a structured, semi-automated pipeline that captures expert radiologist knowledge in a parameterized graph representation. Specifically, it defines 11 clinically significant SOPs as nodes, where each node encodes the parameters of an anatomical plane, while edges represent spatial dependencies and the diagnostic search order followed by radiologists during manual interpretation.

The framework begins by segmenting the auditory ossicles: malleus, incus, and stapes from calibrated CT volumes, providing anatomical anchors for SOP localization. Based on these segmentation-derived masks, a heuristic SOP localization algorithm classifies SOPs into three levels of complexity (standard, single-axis rotation, and multi-axis rotation) and estimates their spatial position through a combination of slice search and axis-specific rotation estimation. This enables accurate detection of both standard and oblique SOPs. The predicted SOP-Graph is then passed through a graph-matching verification step, where it is

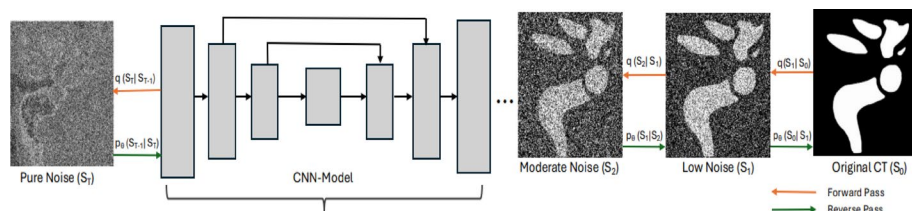


Fig. 6 Illustration of the diffusion-based process for otologic structure segmentation using a representative temporal bone CT image of the middle ear, highlighting the stapes as a key ossicular structure involved in sound transmission. In the forward process $q(S_T|S_{T-1})$ (orange arrows), gaussian noise is progressively added to the original otologic segmentation mask S_0 , ultimately producing a noise-only representation S_T . In the reverse process $p_0(S_{T-1}|S_T)$ (green arrows), a CNN model iteratively removes noise while conditioning on the CT image, refining intermediate noisy masks into an anatomically consistent final segmentation of otologic structures

compared against a radiologist-curated reference graph. This step not only filters out failed annotations but also ensures that the annotated planes maintain proper spatial consistency, are anatomically grounded, and align with diagnostic meaningful views.

Although the SOP-Graph is not a direct segmentation model, it plays a critical supporting role in otologic segmentation workflows. The localized SOP landmarks serve as strong anatomical priorities to guide downstream segmentation models. They allow ROI extraction, narrowing the segmentation focus to clinically relevant sub volumes, which reduces background noise, speeds up processing, and enhances fine structural detail capture. When integrated with architectures such as UNETR or diffusion-based DiffVNet, this improves training efficiency, reduces false positives, and enhances boundary delineation in small, low-contrast targets.

The model was evaluated on a dataset of 610 high-resolution temporal bone CT scans, with an average annotation success rate of 65.8% after radiologist verification. Moreover, it reduced annotation time from approximately 12 min to 2 min per case and enabled the creation of a large SOP-annotated dataset containing 4414 labeled slices. These results demonstrate their effectiveness in improving annotation efficiency, anatomical accuracy, and reproducibility, making it a valuable tool for accelerating dataset preparation and improving segmentation quality in otologic imaging pipelines.

4 Discussion

Motivated by persistent challenges in segmentation accuracy, limited training dataset, and model generalizability, this review systematically evaluates DL-based approaches for otologic structure segmentation developed between 2020 and 2025. Our aim is to analyze advanced architectures targeting the outer ear, middle ear, and inner ear with a particular emphasis on hybrid models designed to meet the unique demands of small and morphologically complex anatomical structures.

Based on the analysis, CNNs (Heutink et al. 2020; Li et al. 2020a; Nikan et al. 2020; Lv et al. 2021; Neves et al. 2021; Hussain et al. 2021; Wang et al. 2021a; Vaidyanathan et al. 2021; Wang et al. 2022a; Ke et al. 2023; Zhou and Li 2023; Stebani et al. 2023; Lim et al. 2024; Ding et al. 2023) remain the most widely adopted backbone for otologic structure

segmentation, reflecting their early dominance and the popularity of U-Net style encoder–decoder frameworks. Over time, CNN-based models have been enhanced through multi-scale skip connections (Li et al. 2023a; Lv et al. 2021; Vaidyanathan et al. 2021; Zhou and Li 2023; Ronneberger et al. 2015) and residual modules (Neves et al. 2021; Zhou and Li 2023), which improve feature representation and preserve fine-grained spatial detail. Skip connections retain anatomical precision and ensure that high-resolution information is passed into the decoding path.

In parallel, hybrid transformer-based models (Li et al. 2023b; Neves et al. 2024) have emerged as a powerful alternative. Unlike CNNs, which primarily capture local spatial features through convolution, Transformers employ self-attention mechanisms to model long-range dependencies across the entire volume. This enables them to capture global semantic context and emphasize clinically relevant regions, particularly useful for small or low contrast otologic structures. vision transformer–based architectures (e.g., UNETR, Swin-UNETR) embed image patches into tokens and process them through multiple attention layers, ensuring that both global context and local spatial detail are effectively integrated. Finally, across both CNN and Transformer-based frameworks, data remains a critical strategy to improve robustness, allowing models to generalize better to unseen clinical cases.

Many of the methods (Li et al. 2023b; Neves et al. 2024) were originally designed for large-organ segmentation and therefore underperform when applied to otologic structures due to ambiguous boundaries, low contrast, and complex spatial distributions. Even with optimized training and high-resolution scans, fine boundary delineation remains challenging. To address these limitations, recent research increasingly integrates transformer-based encoders (Chen et al. 2021), spatial–channel attention modules (Alwadee et al. 2025), and hybrid CNN–transformer architectures (Wang et al. 2022c; Li et al. 2023b) which jointly capture local texture detail and global anatomical context, enabling more precise segmentation of small and complex structures.

State-of-the-art research shows a growing adoption of advanced architectures such as diffusion-based (Xie et al. 2025) and graph-based models (Li et al. 2025b). These approaches highlight a shift from purely performance-driven designs toward methods that also address practical challenges, including limited data availability, annotation effort, and computational efficiency. Diffusion-based frameworks have been applied in semi-supervised settings to alleviate the scarcity of annotated datasets, while graph-based models have been utilized to reduce segmentation time complexity through efficient landmark localization. More recently, both strategies often guided by anatomical priorities have been increasingly integrated with CNN and transformer backbones, accelerating the development of hybrid architectures. This integration reflects the field's active exploration of attention-driven and multi-paradigm designs aimed at overcoming CNN-specific limitations such as restricted receptive fields, limited capacity for modeling long-range dependencies, and vulnerability to small, annotated datasets.

In terms of anatomical targets, the most frequently segmented structures are the cochlea, ossicles, and semicircular canals which represent key components of the inner and middle ear that are essential for hearing and balance. Their well-defined anatomy and high clinical relevance make them indispensable not only for surgical planning but also for disease diagnosis, vestibular disorder evaluation, post-operative assessment, and trauma detection. In contrast, structures such as the facial nerve, vestibule, cochlear canal, stapes, incus, and malleus remain under-explored due to their small size, ambiguous morphology, anatomical

variability, and inconsistent boundaries, which complicate accurate segmentation. Furthermore, segmentation of the outer ear including the external auditory canal and auricle has received relatively limited attention in the literature, despite its importance in conditions such as congenital malformations, external ear trauma, and preoperative planning for reconstructive surgery.

From an evaluation perspective, the DSC remains the most widely reported metric for otologic structure segmentation, largely due to its simplicity and straightforward interpretability. However, its reliance on volumetric overlap makes it less sensitive to subtle boundary deviations that is a critical limitation in otologic imaging where structures such as the facial nerve, ossicles, and stapes are extremely small and tightly clustered. For instance, while ultra-HRCT imaging can yield DSC values approaching 0.90 for the cochlea (Heutink et al. 2020), the facial nerve often falls below 0.70 (Neves et al. 2021; Wang et al. 2021a; Zhou and Li 2023) because of its thin, tortuous course. Similarly, the stapes remains difficult to segment in CT volumes owing to motion artifacts and inherently low contrast (Zhou et al. 2019; Li et al. 2020a; Nikan et al. 2020; Neves et al. 2021). These discrepancies highlight the clinical risk that even minor sub-voxel boundary errors may alter perceived anatomical relationships, with potential consequences for both diagnostic interpretation and surgical planning.

To mitigate these shortcomings, distance-based metrics such as HD and its robust variants (HD95, mHD, aHD) are increasingly reported as complementary measures. These metrics provide valuable sensitivity in edge-critical regions, where precise contour alignment is essential for example, distinguishing ossicles from surrounding bone or tracing the narrow course of the facial nerve. Yet, despite their diagnostic relevance, only a subset of studies in the reviewed literature reports them alongside DSC, pointing to a persistent gap in comprehensive evaluation practices for otologic structure segmentation.

Regarding imaging modalities, CT, particularly HRCT and ultra-HRCT, remains the dominant modality due to its superior spatial resolution for visualizing fine bony anatomy (Xie et al. 2025; Heutink et al. 2020; Wang et al. 2022a). Micro-CT provides even greater detail but is restricted to cadaveric studies (Guo et al. 2024). MRI is rarely used in otologic structure segmentation because of its limited visibility for bony structures, despite offering complementary soft-tissue contrast (Vaidyanathan et al. 2021). The dataset sizes in the reviewed studies range from approximately 100 to 600 patient cases but many are less than 100. However, key acquisition details such as scanner model, voxel size, reconstruction kernel, and imaging protocol are often omitted. This lack of standardized and transparent metadata undermines reproducibility and complicates cross-study performance comparisons. Consequently, models trained on limited or institution-specific data are more prone to overfitting, reducing their generalizability and clinical applicability in multi-center otologic imaging.

Furthermore, all datasets were collected from single institutions, resulting in limited diversity in patient demographics, scanner hardware, and acquisition settings. This homogeneity increases the risk of overfitting to institution-specific imaging characteristics and reduces cross-institution generalizability, thereby limiting the ability of otologic structure segmentation models to perform reliably in multi-center clinical environments. Addressing these issues requires larger, multi-center datasets and streamlined annotation workflows. The SOP-graph study, which processed 640 HRCT scans, demonstrates the feasibility of

reducing annotation time through automated landmark-based frameworks, enabling scalable dataset curation.

Publication activity has steadily increased from 2023 to 2025, driven by the adoption of transformers, diffusion-based models, and hybrid architectures. This shift indicates a departure from conventional CNNs toward more specialized architectures tailored for otologic structure segmentation. In semi-supervised contexts, the diffusion-based model (Xie et al. 2025) is employed not only to address the scarcity of annotated datasets but also to improve learning stability.

Graph-based models (Li et al. 2025b), such as the SOP graph, reduce segmentation complexity by reframing the task as anatomical landmark localization rather than dense voxel-wise labeling. This strategy accelerates inference, lowers computational cost, and streamlines annotation. When integrated into segmentation pipelines through ROI extraction or spatial constraints, graph-derived priors enhance boundary delineation, maintain structural fidelity, and reduce false positives in complex otologic regions.

Despite methodological advances, the number of open-source releases in otologic structure segmentation remains critically low, a fewer than ten identified in the reviewed literature, severely limiting reproducibility, independent validation, and cross-benchmarking. Public release of source code pretrained model weights, and standardized training pipelines would allow other researchers to faithfully replicate reported results, identify potential shortcomings, and extend existing methods rather than duplicating prior work. In the same way, making anonymized datasets available or at minimum providing comprehensive acquisition metadata such as scanner model, voxel dimensions, reconstruction kernel, and imaging protocol would mitigate domain shift risks and support robust multi-center validation.

5 Conclusion and future work

In this survey, we have presented a comprehensive review of DL-based segmentation approaches for the entire otologic anatomy, encompassing the outer ear, middle ear, and inner ear. We identified challenges unique to otologic structures, including their extremely small size, complex morphology, low contrast, and distribution across multiple, spatially scattered components and then systematically reviewed the literature from 2020 to 2025, highlighting how recent works have addressed these challenges through architectural innovations and training strategies. Our analysis traced the evolution from conventional CNN-based architectures to more advanced strategies, including transformer-based, hybrid CNN–transformer models, diffusion-based frameworks, and graph neural networks. For each category, we discussed key technological advancements, application-specific limitations, and emerging trends.

To enable real-world clinical adoption of otologic segmentation systems, future research must address several critical challenges. Although hybrid and transformer-based state-of-the-art models have demonstrated improved accuracy, their high computational complexity limits deployment in resource-constrained healthcare environments. Many clinical settings lack access to high performance graphical processing units, and inference latency remains a concern. Future work should focus on the development of lightweight architectures (Alwadee et al. 2025), model pruning strategies (Fernandes and Yen 2021), and efficient

attention mechanisms (Hassan et al. 2026) (Zhong et al. 2025; Ali and Ghazanfar 2025) that preserve segmentation performance while enabling faster, real-time inference.

Another critical challenge is the limited size and diversity of current datasets. Most existing models are trained on small, single-institution datasets, often with fewer than 100 patient studies, which limits their generalizability to external datasets or pathological cases. The largest publicly reported dataset contains only 325 scans from a single institution, which was still insufficient for robust model validation. Demographic variation is only one facet of diversity; other important factors include differences in anatomical morphology, pathology prevalence, imaging acquisition protocols, reconstruction kernels, voxel sizes, and scanner hardware. This lack of diversity increases the risk of overfitting to institution-specific characteristics, thereby reducing model's adaptability in broader clinical contexts. In otologic structure segmentation, small, low-contrast structures such as the incus, stapes, malleus, facial nerve, and semicircular canals are highly sensitive to resolution and protocol variations, these limitations may result in substantial performance degradation when models are applied outside their training domain. Large-scale, multi-institutional datasets capturing a broad range of anatomical, demographic, and technical variations are therefore essential for developing robust, clinically deployable segmentation systems.

To mitigate domain shifts introduced by variations in imaging protocols, scanner hardware, and patient populations, a range of domain adaptation strategies can be employed. Data-level (Gupta and Gupta 2025) approaches such as style transfer, histogram matching, and resolution harmonization help standardize image appearance across sites while preserving anatomical fidelity. Feature-level methods (Wu et al. 2025a) employ adversarial training or domain-invariant embedding constraints to ensure learned feature representations remain robust across heterogeneous data sources. Output-level strategies (Liu et al. 2025c) leverage shape prior, contour regularization, and uncertainty-aware refinement to maintain anatomical plausibility in the segmentation results.

Although CT remains the dominant imaging modality for otologic structure segmentation due to its high spatial resolution and detailed depiction of bony anatomy, its limited soft-tissue contrast may hinder accurate delineation of structures. MRI, with superior soft-tissue visualization, offers complementary information that could enhance segmentation accuracy. Integrating CT and MRI through DL-based synthesis or feature-level fusion enables combined analysis that leverages the strengths of both modalities. Techniques such as attention-guided fusion networks (Song et al. 2025; Zhu et al. 2025), cross-modality feature transformers (Wang et al. 2025; Wu et al. 2025b), and CT-to-MRI synthesis (Xia et al. 2025; Kim et al. 2025) have demonstrated potential for improved soft-tissue and bony structure differentiation. In otologic applications, these approaches might aid in challenging cases such as identifying the facial nerve within dense bone or detecting labyrinthine fistulas. A unified multi-modal analysis framework could significantly enhance robustness and clinical utility in fine scale otologic structure segmentation.

Finally, the continued reliance on manual annotation for small and intricate otologic structures remains a major constraint, as it is both time-consuming and labor-intensive. Semi-supervised and self-supervised learning methods, including pseudo-labeling (Wang et al. 2022b) contrastive learning (Shu et al. 2024), unsupervised pretraining and transfer learning (Taher et al. 2025; Tozuka et al. 2026), have shown strong potential to reduce annotation demands in other medical imaging domains, and can be readily adopted for otologic structure segmentation. When combined with data augmentation (Sun et al. 2025),

and domain adaptation (Li et al. 2025a; Cai et al. 2025; Li et al. 2025c), these methods can further enhance generalization while minimizing dependence on expert-labeled data.

In summary, addressing challenges related to computational efficiency, data scalability, multi-modal integration, and label-efficient learning will be essential for developing clinically robust, generalizable, and deployable otologic segmentation frameworks. Despite notable methodological progress, clinical applicability is currently limited due to the lack of large-scale external and multi-center validation studies. Most current models have been evaluated on small, single-institution datasets, and their robustness across diverse clinical settings has not been sufficiently demonstrated. Therefore, extensive multi-center validation remains an essential prerequisite to fully justify routine clinical implementation. These insights provide a foundation for future research toward clinically viable systems that may ultimately enhance diagnostic accuracy, guide surgical planning, and improve patient outcomes.

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Data availability No datasets were generated or analysed during the current study.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Abdullah TH (2025) Deep learning in cervical cancer detection: a systematic review. Arch Comput Methods Eng. <https://doi.org/10.1007/s11831-025-10490-7>

- AbuHaweeleh MN et al (2025) Sepsis mortality prediction using machine learning and deep learning-a systematic review. *BMC Med Inform Decis Mak*. <https://doi.org/10.1186/s12911-025-03286-z>
- Ali F, Ghazanfar Z (2025) Enhanced deepfake detection through multi-attention mechanisms: a comprehensive framework for synthetic media identification. *ICCK Trans Intell Syst* 2(4):248–258
- Ali SG et al (2021) Cost-effective broad learning-based ultrasound biomicroscopy with 3D reconstruction for ocular anterior segmentation. *Multimedia Tools Appl* 80(28):35105–35122
- Ali SG et al (2024) AI-enhanced digital technologies for myopia management: Advancements, challenges, and future prospects. *Vis Comput* 40(6):3871–3887
- Ali SG et al (2025) Revolutionizing diabetic retinopathy and macular edema management: a systematic review on the transformative potential of artificial intelligence: SG Ali et al. *Visual Comput* 41(12):10109–10134
- Alwadee EJ, Sun X, Qin Y, Langbein FC (2025) LATUP-Net: a lightweight 3D attention U-Net with parallel convolutions for brain tumor segmentation. *Comput Biol Med* 184:109353
- Arar KH, Özen H, Polat G, Turan S (2025) Artificial intelligence, generative artificial intelligence and research integrity: a hybrid systemic review. *Smart Learn Environ* 12(1):44
- Archana R, Jeevaraj PE (2024) Deep learning models for digital image processing: a review. *Artif Intell Rev* 57(1):11
- Bassiouni M, Bauknecht H-C, Muench G, Olze H, Pohlen J (2023) Missed radiological diagnosis of otosclerosis in high-resolution computed tomography of the temporal bone—retrospective analysis of imaging, radiological reports, and request forms. *J Clin Med* 12(2):630
- Benson JC, Lane JI (2022) Temporal bone anatomy. *Neuroimaging Clin N Am* 32(4):763–775
- Boutillon A, Borotikar B, Burdin V, Conze P-H (2022) Multi-structure bone segmentation in pediatric MR images with combined regularization from shape priors and adversarial network. *Artif Intell Med* 132:102364
- Cai Z et al (2025) Style mixup enhanced disentanglement learning for unsupervised domain adaptation in medical image segmentation. *Med Image Anal* 101:103440
- Chen J et al (2021) Transunet: transformers make strong encoders for medical image segmentation, arXiv preprint arXiv:2102.04306
- Chen T et al (2025) Zig-rir: Zigzag rwkv-in-rwkv for efficient medical image segmentation. *IEEE Trans Med Imaging*. <https://doi.org/10.1109/TMI.2025.3561797>
- Cuellar JR, Gillette L, Dinh V, Woodard P, Burri M, Klingensmith JD (2024) Echocardiogram image segmentation and cardiac adipose tissue estimation using spectral analysis and deep learning. *Medical imaging 2024: ultrasonic imaging and tomography*, vol 12932. SPIE, Bellingham, pp 57–65
- Ding AS et al (2023) A self-configuring deep learning network for segmentation of temporal bone anatomy in cone-beam CT imaging. *Otolaryngology–Head Neck Surg* 169(4):988–998
- Dosovitskiy A et al (2020) An image is worth 16x16 words: transformers for image recognition at scale, arXiv preprint arXiv:2010.11929
- Fausser J et al (2019) Toward an automatic preoperative pipeline for image-guided temporal bone surgery. *Int J Comput Assist Radiol Surg* 14(6):967–976
- Fernandes FE Jr, Yen GG (2021) Pruning of generative adversarial neural networks for medical imaging diagnostics with evolution strategy. *Inf Sci* 558:91–102
- Fu L, Chen Y, Ji W, Yang F (2024) Sstrans-net: smart Swin transformer network for medical image segmentation. *Biomed Signal Process Control* 91:106071
- Gade M, Nguyen KM, Gedde S, Fernandez-Quilez A (2024) Impact of uncertainty quantification through conformal prediction on volume assessment from deep learning-based MRI prostate segmentation. *Insights Imaging* 15(1):286
- Gao Y, Zhou M, Metaxas DN (2021) UTNet: a hybrid transformer architecture for medical image segmentation, In *International conference on medical image computing and computer-assisted intervention*, Springer, pp. 61–71
- Guo X, Lin X, Yang X, Yu L, Cheng K-T, Yan Z (2024) UCTNet: uncertainty-guided CNN-Transformer hybrid networks for medical image segmentation. *Pattern Recognit* 152:110491
- Gao Y, Zhang J, Wei S, Li Z (2025) PFormer: An efficient CNN-Transformer hybrid network with content-driven P-attention for 3D medical image segmentation. *Biomed Signal Process Control* 101:107154
- Gupta S, Gupta S (2025) Feature extraction and feature selection procedures for medical image analysis. *Computer-Assisted Analysis for Digital Medicinal Imagery*. IGI Global Scientific Publishing, pp 221–280
- Hasan Z et al (2023) Convolutional neural networks in ENT radiology: systematic review of the literature. *Annals Otolaryngology Rhinology Laryngology* 132(4):417–430
- Hassan MZ, Gazis A, Khan A, Ghazanfar Z (2026) Learning cross-modal collaboration via pyramid attention for RGB thermal sensing in saliency detection. *ICCK Trans Sens Communication Control* 3(1):1–14

- Heutink F et al (2020) Multi-Scale deep learning framework for cochlea localization, segmentation and analysis on clinical ultra-high-resolution CT images. *Comput Methods Programs Biomed* 191:105387
- Hussain R, Lalande A, Girum KB, Guigou C, Bozorg Grayeli A (2021) Automatic segmentation of inner ear on CT-scan using auto-context convolutional neural network. *Sci Rep* 11(1):4406
- Iqbal S, Khan TM, Naqvi SS, Naveed A, Meijering E (2025) TBConvL-Net: a hybrid deep learning architecture for robust medical image segmentation. *Pattern Recognit* 158:111028
- Isensee F, Jaeger PF, Kohl SA, Petersen J, Maier-Hein KH (2021) nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nat Methods* 18(2):203–211
- Kashyap M et al (2025) Automated deep learning-based detection and segmentation of lung tumors at ct imaging. *Radiology* 314(1):e233029
- Ke J et al (2023) Deep learning-based approach for the automatic segmentation of adult and pediatric temporal bone computed tomography images. *Quant Imaging Med Surg* 13(3):1577
- Kim DY, Oh HW, Suh CH (2023) Reporting quality of research studies on AI applications in medical images according to the CLAIM guidelines in a radiology journal with a strong prominence in Asia. *Korean J Radiol* 24(12):1179
- Kim KH, Lee E-C, Yoon YD, Shin D-W, Koo H-W, Lee B-J (2025) Translation of computed tomography images to T2-Weighted magnetic resonance images of lumbar spine using generative adversarial networks on sagittal images. *Sci Rep* 15(1):18385
- Lan L, Cai P, Jiang L, Liu X, Li Y, Zhang Y (2026) Brau-net++: U-shaped hybrid cnn-transformer network for medical image segmentation. *IEEE Trans Radiation Plasma Med Sci*. <https://doi.org/10.1109/TRPMS.2026.3666783>
- Lee JW et al (2024) Variability in manual segmentation of temporal bone structures in cone beam CT images. *Otol Neurotol* 45(3):e137–e141
- Li X, Gong Z, Yin H, Zhang H, Wang Z, Zhuo L (2020a) A 3D deep supervised densely network for small organs of human temporal bone segmentation in CT images. *Neural Netw* 124:75–85
- Li M, Wang C, Zhang H, Yang G (2020b) MV-RAN: multiview recurrent aggregation network for echocardiographic sequences segmentation and full cardiac cycle analysis. *Comput Biol Med* 120:103728
- Li X et al (2021) Dual attention deep fusion semantic segmentation networks of large-scale satellite remote-sensing images. *Int J Remote Sens* 42(9):3583–3610
- Li C, Li X, Zhou R (2023a) Cochlear CT image segmentation based on u-net neural network. *J Radiat Res Appl Sci* 16(2):100560
- Li Z, Zhou L, Tan S, Tang A (2023b) Application of UNETR for automatic cochlear segmentation in temporal bone CTs. *Auris Nasus Larynx* 50(2):212–217
- Li X et al (2023c) A synergistical attention model for semantic segmentation of remote sensing images. *IEEE Trans Geosci Remote Sens* 61:1–16
- Li L, Liu J, Xiao H, Zhou G, Liu Q, Zhang Z (2025a) Expert guidance and partially-labeled data collaboration for multi-organ segmentation. *Neural Netw* 187:107396
- Li X et al (2025b) Standard observation plane annotation for diagnosis of ossicular chain in the temporal bone CT images. *Med Biol Eng Comput* 63(6):1683–1695
- Li H et al (2025c) AIF-SFDA: autonomous information filter driven source-free domain adaptation for medical image segmentation, In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 5, pp. 4716–4724
- Lim J, Abily A, Ben Salem D, Gaillandre L, Attie A, Ognard J (2024) Training and validation of a deep learning U-net architecture general model for automated segmentation of inner ear from CT. *Eur Radiol Exp* 8(1):104
- Lingam RK, Kumar R, Vaidhyath R (2019) Inflammation of the temporal bone. *Neuroimaging Clin N Am* 29(1):1–17
- Liu Z et al (2021) Swin transformer: Hierarchical vision transformer using shifted windows, In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10012–10022
- Liu X et al (2025a) Directional latent space representation for medical image segmentation. *Visual Comput* 41(5):3095–3107
- Liu Z et al (2025b) Dilated dendritic learning of global–local feature representation for medical image segmentation. *Expert Syst Appl* 264:125874
- Liu Q, Gu X, Henderson P, Dai H, Deligianni F (2025c) Certainty-guided cross contrastive learning for semi-supervised medical image segmentation. *IEEE Trans Biomed Eng*. <https://doi.org/10.1109/TBME.2025.3541992>
- Lv Y, Ke J, Xu Y, Shen Y, Wang J, Wang J (2021) Automatic segmentation of temporal bone structures from clinical conventional CT using a CNN approach. *Int J Med Robot Comput Assist Surg* 17(2):e2229
- Manzari ON, Kaleybar JM, Saadat H, Maleki S (2024) BEFUnet: A hybrid CNN-transformer architecture for precise medical image segmentation, arXiv preprint arXiv:2402.08793

- Masood S et al (2024) Deep choroid layer segmentation using hybrid features extraction from OCT images. *Visual Comput* 40(4):2775–2792
- Minace S, Boykov Y, Porikli F, Plaza A, Kehtarnavaz N, Terzopoulos D (2021) Image segmentation using deep learning: A survey. *IEEE Trans Pattern Anal Mach Intell* 44(7):3523–3542
- Mostafa AM, Zakariah M, Aldakheel EA (2023) Brain tumor segmentation using deep learning on MRI images. *Diagnostics* 13(9):1562
- Müller D, Kramer F (2021) MIScnn: a framework for medical image segmentation with convolutional neural networks and deep learning. *BMC Med Imaging* 21(1):12
- Natour A, Doyle E, DeDio R, Samy RN (2024) Temporal bone fracture related facial palsy: efficacy of decompression with and without grafting. *Curr Opin Otolaryngol Head Neck Surg* 32(5):294–300
- Neves C, Tran E, Kessler I, Blevins N (2021) Fully automated preoperative segmentation of temporal bone structures from clinical CT scans. *Sci Rep* 11(1):116
- Neves CA, Chemaly TE, Fu F, Blevins NH (2024) Deep learning method for rapid simultaneous multistructure temporal bone segmentation. *Otolaryngol Head Neck Surg* 170(6):1570–1580
- Nikan S et al (2020) PWD-3DNet: a deep learning-based fully-automated segmentation of multiple structures on temporal bone CT scans. *IEEE Trans Image Process* 30:739–753
- Niyas S, Pawan S, Kumar MA, Rajan J (2022) Medical image segmentation with 3D convolutional neural networks: a survey. *Neurocomputing* 493:397–413
- Nogovitsyn N et al (2019) Testing a deep convolutional neural network for automated hippocampus segmentation in a longitudinal sample of healthy participants. *Neuroimage* 197:589–597
- O'Shea RJ, Sharkey AR, Cook GJ, Goh V (2021) Systematic review of research design and reporting of imaging studies applying convolutional neural networks for radiological cancer diagnosis. *Eur Radiol* 31(10):7969–7983
- Page MJ et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. <https://doi.org/10.1136/bmj.n71>
- Pham N et al (2022) High-resolution CT imaging of the temporal bone: a cadaveric specimen study. *J Neurol Surg B Skull Base* 83(05):470–475
- Powell KA, Kashikar T, Hittle B, Stredney D, Kerwin T, Wiet GJ (2019) Atlas-based segmentation of temporal bone surface structures. *Int J Comput Assist Radiol Surg* 14(8):1267–1273
- Ronneberger O, Fischer P, Brox T (2015) U-net: convolutional networks for biomedical image segmentation, In *International Conference on Medical image computing and computer-assisted intervention*, Springer, pp. 234–241
- Ross T, Tanna R, Lilaonitkul W, Mehta N (2024) Deep learning for automated image segmentation of the middle ear: A scoping review. *Otolaryngol Head Neck Surg* 170(6):1544–1554
- Sahoo AK, Parida P, Muralibabu K, Dash S (2023) Efficient simultaneous segmentation and classification of brain tumors from MRI scans using deep learning. *Biocybern Biomed Eng* 43(3):616–633
- Salimi Y, Shiri I, Mansouri Z, Zaidi H (2025) Development and validation of fully automated robust deep learning models for multi-organ segmentation from whole-body CT images. *Phys Med* 130:104911
- Shu C et al (2024) Miter: Medical image–text joint adaptive pretraining with multi-level contrastive learning. *Expert Syst Appl* 238:121526
- Si L et al (2022) Deep learning in knee imaging: a systematic review utilizing a Checklist for Artificial Intelligence in Medical Imaging (CLAIM). *Eur Radiol* 32(2):1353–1361
- Singh A et al (2025) Deep learning based coronary artery disease detection and segmentation using ultrasound imaging with adaptive gated SCNN models. *Biomed Signal Process Control* 105:107637
- Song M, Zhai C, Yang L, Liu Y, Bian G (2025) An attention-guided multi-scale fusion network for surgical instrument segmentation. *Biomed Signal Process Control* 102:107296
- Stebani J, Blaimer M, Zabler S, Neun T, Pelt DM, Rak K (2023) Towards fully automated inner ear analysis with deep-learning-based joint segmentation and landmark detection framework. *Sci Rep* 13(1):19057
- Strudel R, Garcia R, Laptev I, Schmid C (2021) Segmenter: transformer for semantic segmentation, In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 7262–7272
- Sun D, Dornaika F, Barrena N (2025) HSMix: hard and soft mixing data augmentation for medical image segmentation. *Inf Fusion* 115:102741
- Swathi N, Sankaran KS (2025) Advances and clinical implications of neuroanatomical biomarkers in schizophrenia by optimizing brain tissue segmentation. *Biomed Signal Process Control* 102:107215
- Taher MRH, Haghighi F, Gotway MB, Liang J (2025) Large-scale benchmarking and boosting transfer learning for medical image analysis. *Med Image Anal* 102:103487
- Tang H et al (2024) HTC-Net: A hybrid CNN-transformer framework for medical image segmentation. *Biomed Signal Process Control* 88:105605
- Tejani AS et al (2023) Updating the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) for reporting AI research. *Nat Mach Intell* 5(9):950–951

- Tozuka R et al (2026) Fractal-driven self-supervised learning enhances early-stage lung cancer GTV segmentation: a novel transfer learning framework. *Japanese J Radiol* 44(1):183–195
- Tricco AC et al (2018) PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med* 169(7):467–473
- Tsuno NS et al (2022) Imaging the external ear: practical approach to normal and pathologic conditions. *Radiographics* 42(2):522–540
- Vaidyanathan A et al (2021) Deep learning for the fully automated segmentation of the inner ear on MRI. *Sci Rep* 11(1):2885
- Wang J et al (2021a) Fully automated segmentation in temporal bone CT with neural network: a preliminary assessment study. *BMC Med Imaging* 21(1):166
- Wang W et al (2021b) Pyramid vision transformer: a versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 568–578
- Wang X-R et al (2022a) Application value of a deep learning method based on a 3D V-Net convolutional neural network in the recognition and segmentation of the auditory ossicles. *Front Neuroinform* 16:937891
- Wang Y et al (2022b) Semi-supervised semantic segmentation using unreliable pseudo-labels. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4248–4257
- Wang J, Huang Q, Tang F, Meng J, Su J, Song S (2022c) Stepwise feature fusion: local guides global. In *International conference on medical image computing and computer-assisted intervention*, Springer, pp. 110–120
- Wang W, Pan B, Ai Y, Li G, Fu Y, Liu Y (2024a) LightCM-PNet: a lightweight pyramid network for real-time prostate segmentation in transrectal ultrasound. *Pattern Recogn* 156:110776
- Wang X et al (2024b) DSIS-DPR: Structured instance segmentation and diffusion prior refinement for dental anatomy learning. *IEEE Trans Multimedia* 26:9464–9476
- Wang M, Chen H, Mao L, Jiao W, Han H, Zhang Q (2025) C5-net: cross-organ cross-modality cswin-transformer coupled convolutional network for dual task transfer learning in lymph node segmentation and classification. *Comput Med Imaging Graph*. <https://doi.org/10.1016/j.compmedimag.2025.102633>
- Wu J, Ji W, Fu H, Xu M, Jin Y, Xu Y (2024) Medsegdiff-v2: diffusion-based medical image segmentation with transformer. In *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, no. 6, pp. 6030–6038
- Wu Z, Xue B, Zhang M (2025a) Multitree gp-based feature learning for multimodal medical image classification. *IEEE congress on evolutionary computation (CEC)*. IEEE, New York City, pp 1–8
- Wu Y, Yuan H, Yuan Z, Meng L, Bai Y, Wang H (2025b) An asymmetric siamese transformer with global-local alignment attention for visible-X-Ray cross-modality package re-identification. *IEEE Trans Inf Forensics Secur*. <https://doi.org/10.1109/tifs.2025.3592540>
- Xia YF et al (2025) Region-guided focal adversarial learning for CT-to-MRI translation: a proof-of-concept and validation study in hepatocellular carcinoma. *Med Phys* 52(5):2861–2873
- Xie M et al (2025) Artificial intelligence model for automatic 3-dimensional reconstruction of ossicular chain and bony labyrinth from high-resolution CT. *Radiol Adv* 2(1):umaf004
- Xu L, Halike A, Sen G, Sha M (2025) Medical image segmentation model based on local enhancement driven global optimization. *Sci Rep* 15(1):18281
- Zhang Y, Liu H, Hu Q (2021) Transfuse: fusing transformers and cnns for medical image segmentation. In *International conference on medical image computing and computer-assisted intervention*, Springer, pp. 14–24
- Zhong J et al (2022) An updated systematic review of radiomics in osteosarcoma: utilizing CLAIM to adapt the increasing trend of deep learning application in radiomics. *Insights Imaging* 13(1):138
- Zhong J et al (2025) PMFSNet: Polarized multi-scale feature self-attention network for lightweight medical image segmentation. *Comput Methods Programs Biomed* 261:108611
- Zhou Z, Siddiquee MMR, Tajbakhsh N, Liang J (2019) Unet++: redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Trans Med Imaging* 39(6):1856–1867
- Zhou L, Li Z (2023) Automatic multi-label temporal bone computed tomography segmentation with deep learning. *Int J Med Robot* 19(5):e2536
- Zhu X, Li S, Bi H, Guan L, Liu H, Lu Z (2025) Automatic choroid segmentation and thickness measurement based on mixed attention-guided multiscale feature fusion network. *IEEE Trans Med Imaging*. <https://doi.org/10.1109/TMI.2025.3597026>

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